

Mohammad Nour

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PROFESSIONAL SUMMARY

Third-year PhD researcher specializing in hardware security and machine learning for cryptographic analysis. Experienced in side-channel attacks, deep learning for key recovery, and explainable AI. Strong background in cryptography and ML, with hands-on expertise in PyTorch, TensorFlow, and scikit-learn.

SKILLS

Programming Languages: Verilog, Python, C/C++, MATLAB, Shell scripting, solidity.

DL/ML: PyTorch, PyTorch Lightning; TensorFlow (v1/v2); OpenCV; scikit-learn; NumPy, Pandas.

Platforms & Tooling: Debian/GNU Linux, Windows; Git, Docker, bash; basic CUDA.

PROJECTS

Uncertainty Estimation in Neural Network-enabled Side-channel Analysis

Conference on Cryptographic Hardware and Embedded Systems (CHES), 2026

- Developed a framework that quantifies predictive uncertainty in neural-network side-channel attacks, separating contributions from model behavior and data quality.

There's Waldo: PCB Tamper Forensic Analysis using Explainable AI on Impedance Signatures

IEEE International Symposium on Electromagnetic Compatibility, Signal and Power Integrity (EMC+SIPI), 2025

- Developed an explainable machine-learning framework that uses impedance (S-parameter) signatures to detect and classify PCB tampering with 96% accuracy.

Too hot to be true: Temperature calibration for NN-assisted side-channel analysis

Constructive Approaches for SeCurity Analysis and Design of Embedded systems (CASCADE), 2026

- Studied the effects of temperature calibration on deep learning based SCA.

Swarm in EM Hay: Particle Swarm-guided Probe Placement for EM SCA

USENIX WOOT Conference on Offensive Technologies, 2026

- Studied the effects of temperature calibration on deep learning based SCA.

SOCIAL ACTIVITIES AND LEADERSHIP

- Leading the Iranian Student Association at Worcester Polytechnic Institute (2023–2025), supporting 42 students during a period of geopolitical stress
- Founded and organized the OPTIMIST PhD forum on AI and side-channel analysis, hosting sessions with 20 participants from academia and industry.
- **Mentoring four undergraduate researchers** and promoted reproducibility by releasing six open-source GitHub repositories supporting research artifacts.

AWARDS AND HONERS

- Recipient of the **DAC Young Fellow Award 2025**.
- Invited talk at **New England Hardware Security Day 2025**, MIT.
- **3rd Place Best Poster Award** at New England Hardware Security Day 2024, WPI.

EDUCATION

Worcester Polytechnic Institute

2023 - Present

PhD - Electrical and Computer Engineering | Advisor: Prof. Fatemeh Ganji

Worcester, MA

Courses: Applied cryptography & physical attacks, Physical Security of Microelectronic Systems

Sharif University of Technology

2020 - 2023

M.Sc. - secure communications and cryptography

Courses: Deep learning, Cryptography, foundations of the blockchains